

Yahya Ibrahim, PhD

AI Engineer | Machine
Learning Engineer |
Computer Vision
Developer



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PROFESSIONAL SUMMARY

AI Engineer with a PhD in Computer Vision and 6+ years of experience building and deploying production-grade deep learning systems at HUN-REN SZTAKI. Specialized in 3D vision, generative models, and large-scale ML pipelines. Strong background in: - End-to-end ML system design (data → training → evaluation → deployment) - 3D scene reconstruction (NeRF, 3D Gaussian Splatting, point cloud completion) - Generative models (GANs, inpainting) - Autonomous driving and remote sensing pipelines - HPC-based training and scalable experimentation Author of 11 research papers with 163 citations and h-index of 6, with experience translating research into practical, open-source, and demonstrable systems. Currently seeking AI Engineer / ML Engineer / Computer Vision Developer roles in industry.



PROFESSIONAL EXPERIENCE

HUN-REN SZTAKI – Machine Perception Laboratory

AI/ML Engineer & Researcher

2017 – Present | Budapest, Hungary

Designed, implemented, and deployed deep learning systems for 3D vision, generative modeling, and autonomous systems. Key Engineering Contributions:

- Developed a 3D point cloud completion system (MVPCC-Net) in PyTorch for autonomous driving applications; released as open-source and published in Image and Vision Computing (Elsevier).
- Built a GAN-based blind image inpainting pipeline for heritage restoration (39 citations).
- Implemented end-to-end training pipelines on HPC clusters with experiment tracking and reproducibility, with live online demos.
- Contributed to research-to-production workflow, including demos and deployable prototypes.



EDUCATION

PhD in Computer Vision

PPKE ITK

2017 – 2023 | Budapest, Hungary

Deep Learning-Based Methods on 2D Image and 3D Structure Inpainting

MSc in Mechatronics Engineering

Tishreen University

2013 – 2017 | Lattakia, Syria

Design of 3D Virtual Environment Building System by RGB-D Camera on Vehicle

BSc in Mechatronics Engineering

Tishreen University

2009 – 2013 | Lattakia, Syria



TECHNICAL SKILLS

Programming

Python (Advanced), C++, MATLAB

ML / DL Frameworks

PyTorch, TensorFlow, Keras

Deep Learning & Architectures

Transformers, CNNs, GANs, Autoregressive, VLM, LLM.

3D Vision & Computer Vision

NeRF, 3D Gaussian Splatting, Point Cloud Processing, OpenCV, Open3D, PCL

Infrastructure & Tools

ROS2, Docker, Git, Weights & Biases, HPC/Linux, LaTeX



SELECTED PROJECTS

3D Point Cloud Completion for Autonomous Driving [↗](#)

- Built deep neural network models to reconstruct incomplete LiDAR point clouds.
- Implemented multi-view learning strategy to improve geometric consistency.
- Designed scalable data preprocessing and training pipelines.

GAN-Based Image Inpainting System [↗](#)

Segmentation, Inpainting, GAN

- Developed a novel machine learning-based fully automatic approach for the semantic analysis and documentation of masonry wall images.
- Implemented a training workflow and evaluation pipeline.
- Achieved strong research and citation impact (39 citations).
- Live demo: <https://imgproc.mplab.sztaki.hu/> [↗](#)

AI-based Fusion of Satellite Data for Biodiversity Change Characterization [↗](#)

Remote Sensing · Multispectral Data Fusion

- Designed and implemented a production-oriented AI pipeline for multi-source satellite data fusion and biodiversity change analysis.
- Demonstrates the practical deployment of scalable remote sensing AI pipelines beyond research prototypes. **Live demo:** <https://biodiversity.sztaki.hu/> [↗](#)



RESEARCH IMPACT

- 11 peer-reviewed publications
- 163 citations
- h-index: 6
- i10-index: 5